

Sachdev-Ye-Kitaev model: Holographic principles of fermions and charged black holes

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One can argue that the most valuable breakthroughs in physics normally arises from the exchange of ideas between two distinct fields. The Sachdev-Ye-Kitaev (SYK) model is one example of such phenomena, where the study of a lattice fermionic system can lead to new insights at thermodynamic properties of charged black holes. This is primarily due to the fact that the SYK model has an extensive zero temperature entropy and that its low-energy dynamics cannot be described by particle like description, which are also core features of charged black holes. This essay aims to explore such duality, while also commenting on several other applications of the theory at hand.

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I. INTRODUCTION

The first step towards the discovery of the SYK model was made by Subir Sachdev and Jinwu Ye at [1], where they explore the effects of random interactions on the Heisenberg model. Thereafter, it was noticed by Subir Sachdev that certain mean-field gapless phase are not compatible with a particle-like description, and that a duality of these critical quantum states of matter could be established with the near horizon physics of $AdS_2 \times \mathbb{R}^2$ Reissner-Nordström black holes [2]. The current understanding of these correspondences at the quantum level was then formulated by Alexei Kitaev at the seminal lectures [3, 4], where the SYK model as it is understood nowadays was first presented. Currently, both condensed matter and high energy physics communities have drawn their attention to such critical models and their equivalences.

Perhaps the most intriguing aspect of the SYK model is its extensive zero temperature entropy $S(T \rightarrow 0) \sim N s_0$, despite the absence of a large ground states degeneracy. A feature also shared by charged black holes. Consequently, the gap between two nearby energy eigenstates must be exponentially suppressed, while the density of states should behave as $\rho(E) \sim e^{N s_0}$. This result has

a direct connection with the Planckian dynamics manifested by the system, where the information stored at a given state is quickly scrambled around the Hilbert space. An effect that has been widely explored in the theory of quantum chaos under the name of *fast scramblers*.

As we will discuss, this residual zero temperature entropy can be traced back to that fact that the deep IR has an emergent reparametrization symmetry, which is broken down to $SL(2, \mathbb{R})$ as we consider perturbations of this fixed point. Due to this symmetry breaking mechanism, namely $\text{Diff}(S^1) \rightarrow SL(2, \mathbb{R})$, pseudo-Goldstone excitations (known as soft modes) can propagate in the low-energy regime. This observation ultimately leads to a dense number of states mentioned previously.

II. SYK MODEL

The SYK model is considered in the literature as the simplest of its kind, and is defined as a fermionic system supported on a lattice with no apparent ordering (See figure 1)

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad i, j = 1, \dots, N, \quad (1)$$

$$(2)$$

$$H = \frac{1}{(2N)^{3/2}} \sum_{ijkl} J_{ijkl} c_i c_j c_k c_l. \quad (3)$$

Consequently, the theory is thought to be an usual many-body quantum mechanics problem. Additionally, the coupling constants J_{ijkl} describe a random all-to-all interactions following a normal distribution with expectation values

$$\overline{J_{ijkl}} = 0, \quad \overline{J_{ijkl}^2} = J^2. \quad (4)$$

As a result, in the large N limit *melon diagrams* (See figure 2) become the most predominant correction to the

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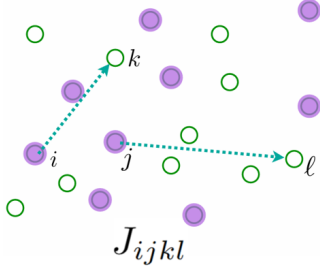


FIG. 1. Example of lattice arrangement of SYK model with half filling and transition amplitudes J_{ijkl} [5].

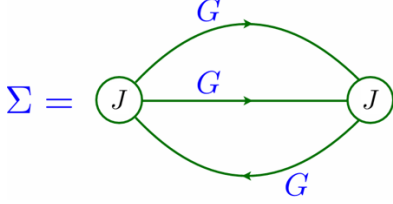


FIG. 2. Melon diagram contained in the self-energy of (3) on the large N limit [5].

propagator $G(\tau_1 - \tau_2)$ of the model

$$G(\tau) = \frac{1}{N} \left\langle \sum_i c_i(\tau) c_i^\dagger(0) \right\rangle \quad (5)$$

$$G^{-1}(i\omega) = -i\omega - \Sigma(i\omega), \quad (6)$$

$$\Sigma(\tau) = -J^2 G(\tau)^3, \quad (7)$$

where τ denotes the Euclidean time, with the identification $\tau \sim \tau + \beta$, and $\Sigma(\tau_1 - \tau_2)$ is the self-energy.

Notice, however, that as we follow to the IR limit, i.e. $\omega \rightarrow 0^+$, the following relation become valid

$$G(i\omega)\Sigma(i\omega) = -1 \quad (8)$$

$$\rightarrow \int d\tau G(\tau_1, \tau) \Sigma(\tau, \tau_2) = -\delta(\tau_1 - \tau_2). \quad (9)$$

Subsequently, we can verify that the theory develops an emergent invariance under reparametrization. Namely, provide $\tau \rightarrow f(\tau)$ we must have

$$G(\tau_1, \tau_2) \rightarrow (f'(\tau_1)^\Delta f'(\tau_2)^\Delta) G(f(\tau_1), f(\tau_2)), \quad (10)$$

$$\Sigma(\tau_1, \tau_2) \rightarrow (f'(\tau_1)^{1-\Delta} f'(\tau_2)^{1-\Delta}) \Sigma(f(\tau_1), f(\tau_2)), \quad (11)$$

with $\Delta = 1/4$. This signals that the low energy physics is described by a conformal field theory and, therefore, is incompatible with a particle-like description given that the correlation length of the model is expected to diverge. Furthermore, provided that we have determined the conformal weight, i.e. $\Delta = 1/4$, it follows that the green function is constrained by the conformal invariance to be

$$G_{CFT}(\tau) = b \frac{\text{sgn}(\tau)}{|\tau|^{2\Delta}}, \quad (12)$$

where b is a constant determined by plugging the ansatz (12) back into (9) and (7)

$$J^2 b^4 = \left(\frac{1}{2} - \Delta \right) \frac{\tan(\pi\Delta)}{\pi}. \quad (13)$$

Nevertheless, the solution $G_{CFT}(\tau)$ is in fact only invariant under the subset $SL(2, \mathbb{R})$ of the allowed reparametrization transformations, that is

$$\tau \rightarrow f(\tau) = \frac{a\tau + b}{c\tau + d}, \quad ad - bc = 1. \quad (14)$$

This indicates the presence of a symmetry breaking mechanism happening at the low energy physics of (3) as we follow slightly away from $\omega \rightarrow 0^+$.

To further inspect this fact, the effective action of the model must be considered, such that it can be proven to have the form

$$Z = \int \mathcal{D}\Sigma \mathcal{D}G e^{N S_{eff}}, \quad S_{eff} = S_{CFT} + S_{soft} \quad (15)$$

$$S_{CFT} = -\frac{1}{2} \log \det (-\Sigma(\tau, \tau')) \quad (16)$$

$$+ \frac{1}{2} \int d\tau d\tau' \left(\Sigma(\tau, \tau') G(\tau, \tau') - \frac{J^2}{4} G(\tau, \tau')^4 \right)$$

$$S_{soft} = -\frac{1}{2} \int d\tau d\tau' \delta(\tau - \tau') \partial_\tau G(\tau, \tau'), \quad (17)$$

Intuitively one can think about $S_{eff} = S_{CFT} + S_{soft}$ as the action that under δG and $\delta \Sigma$ variations reproduces (6) and (7), respectively. See [5, 6] for the explicit derivation of (15).

Naively, it seems reasonable to disregard (17) in the IR limite, given its non-invariance under reparametrizations (10) and (11). Nonetheless, the presence of delta function at S_{soft} yields to a conformal correction to the green function of the model as we perform perturbations around $\tau_+ = (\tau_1 + \tau_2)/2$, that is

$$G(\tau < 1) \approx b \frac{\text{sgn}(\tau)}{|\tau|^{2\Delta}} \left(1 + \tau^2 \frac{\Delta}{6} \text{Sch}(f(\tau), \tau) \right), \quad (18)$$

$$\text{Sch}(f(\tau), \tau) = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2 \quad (19)$$

where $\text{Sch}(f(\tau), \tau)$ is the Schwarzian derivative. In order to derive (18) we have used (10) in combination with

$$|f(\tau_1) - f(\tau_2)| \approx f'(\tau_+) \tau_{12} + f'''(\tau_+) \frac{\tau_{12}^3}{24} \quad (20)$$

$$f'(\tau_1) f'(\tau_2) \approx f'(\tau_+)^2 - (f'(\tau_+) f''(\tau_+))' \frac{\tau_{12}^2}{4}, \quad (21)$$

where $\tau_{12} = \tau_1 - \tau_2$. Accordingly, the effective action must acquire a term of the form

$$S_{sch} = -\frac{C}{J} \int d\tau \left(\frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2 \right) \subset S_{soft}, \quad (22)$$

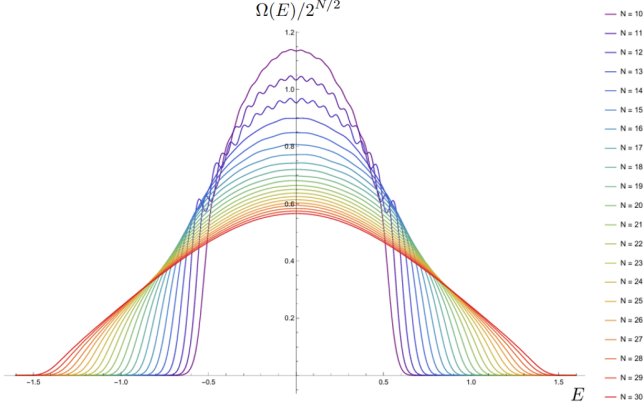


FIG. 3. Normalized finite density of states for the SYK model with $N = 10, \dots, 30$. [7].

which explicitly breaks the reparametrization invariance to the global symmetry group $SL(2, \mathbb{R})$. This result reveals the presence of zero-modes propagating in the low-energy physics, referred to as soft-modes. Consequently, the energy spectrum must host a large number of states in the IR limit given the size of $\text{Diff}(S^1)$. In (22) the constant $C \approx 0.396\dots$ [8] is related to the heat capacity of the model and can only be determined numerically.

With the effective action at hand, we can proceed to compute the lead contributions to the partition function and subsequently the entropy. For that we exploit the reparametrization invariance to construct a finite temperature green function

$$\tau \rightarrow f(\tau) = \tan\left(\frac{\pi(\tau + \epsilon(\tau))}{\beta}\right) \quad (23)$$

$$G_{CFT}(\tau) \rightarrow G_\beta(\tau) = b \left[\frac{\pi}{\beta \sin\left(\frac{\pi\tau}{\beta}\right)} \right]^{2\Delta} \quad (24)$$

and determine the free energy by considering the on-shell effective action $\beta F = S_{eff}[G_\beta(\tau)]$, while taking into account the Schwarzian term, which will essentially capture the temperature dependence of the model. As a result, the partition function becomes

$$Z(\beta) = e^{N s_0} \int \frac{\mathcal{D}f}{|SL(2, \mathbb{R})|} \exp\left(\frac{NC}{2J} \int_0^\beta \left[(\epsilon'')^2 - \left(\frac{2\pi}{\beta}\right)^2 (\epsilon')^2 \right] \right), \quad (25)$$

where the $\epsilon(\tau)$ expansion was carried out on the Schwarzian term, and that under proper regularization the conformal contribution (16) turns out to be the universal constant [5, 8]

$$s_0 = \frac{\mathcal{G}}{2\pi} + \frac{\ln(2)}{8} \approx 0.23242384958\dots, \quad (26)$$

where $\mathcal{G}$ is the Catalan constant.

Notice, however, that due to the presence of the remnant $SL(2, \mathbb{R})$ invariance, we must be careful while integrating over conformal manifold, as zero-modes could lead to divergences. In order to avoid such a problem, we mod out these degrees of freedom from the functional integral by considering the following Fourier decomposition

$$\epsilon(\tau) = \sum_{n \neq 0, \pm 1} a_n e^{\frac{2\pi i n}{\beta} \tau}, \quad (27)$$

such that the quadratic theory given by $\epsilon(\tau)$ results in the final partition function

$$Z(\beta) = \frac{1}{\sqrt{N}} e^{N s_0} \left(\frac{1}{\beta J}\right)^{3/2} \sinh\left(\frac{4\pi^2 N C}{J\beta}\right). \quad (28)$$

Thereby leading to the following entropy and density of states for $N \rightarrow \infty$

$$S(\beta) = N \left(s_0 + \frac{4\pi^2 C}{J\beta} \right) - \frac{3}{2} \ln(\beta J) - \frac{1}{2} \ln(N), \quad (29)$$

$$\Omega(E) = e^{S(E)} \sim \sqrt{\frac{E}{N}} e^{N s_0 + \sqrt{2\gamma N E}}, \quad (30)$$

where $\gamma = 2\pi^2 C/J$.

Having established the thermodynamical properties of the SYK model (29) and (30), we observe the manifestation of a residual entropy in the limit

$$\lim_{T \rightarrow 0} \lim_{N \rightarrow \infty} \frac{S(\beta)}{N} = s_0. \quad (31)$$

Notably, this result should be contrasted with the similar but trivial statement that occurs in extensively large degenerated ground states, where the limits in (31) are taken at opposite order. In fact, the origin of such residual entropy arises from the observation that (30) indicates the presence of a significantly large number of states at the low-energy physics being generated by the soft-modes, as discussed previously. In figure 3 we show the numerical simulation for the density of states performed at [7].

III. THERMODYNAMICS OF BLACK HOLES

With the SYK model now fully solved, we briefly turn to the gravitational side of the duality.

The discovery by Stephen Hawking [9] that black holes should emit radiation and, therefore, possess a temperature inversely proportional to their mass

$$T = \frac{\hbar c^3}{8\pi G k_B} \frac{1}{M}, \quad (32)$$

stands as one of the most profound contributions to modern physics in the last century. As a result of this radiation, black holes are supposed to be unstable and evaporate over time, until they cease to exist.

Furthermore, it can also be inferred that the entropy of a black hole is determined solely by its surface area and charge

$$S_{BH} = \frac{k_B}{4G\hbar} \mathcal{A}(M, Q), \quad (33)$$

which, upon calculating quantum corrections similar to those used previously in the SYK model, lead to [10]

$$S(\beta) = \frac{k_B c^3}{4\hbar G} \mathcal{A} \left(1 + \frac{2\sqrt{\pi\mathcal{A}}}{\beta\hbar c} \right) - \frac{3}{2} \ln \left(\beta \sqrt{\hbar c^5/G} \right) - \frac{559}{180} \ln \left(\frac{k_B c^3}{\hbar G} \mathcal{A} \right). \quad (34)$$

Building on this result, a direct correspondence with the entropy of the SYK model (29) can be established, where the term $k_B c^3 \mathcal{A}/4\hbar G$ display the same role as N . This equivalence therefore implies the existence of an holographic correspondence principle between the near horizon of black hole thermodynamics and the SYK model.

IV. CONCLUSION

In this work we have fully solved the celebrated SYK model, where the emergence of the reparametrization invariance in the deep IR limit, and later its reduction to the $SL(2, \mathbb{R})$ group, have lead to the discovery of propagating zero-modes in the low energy physics. Thus implying in the observation of a non-trivial residual zero temperature entropy. Furthermore, we briefly discussed the correspondence from these results with the thermodynamics of black holes, which is completely captured by their near horizon regime.

V. ACKNOWLEDGMENTS

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- [1] S. Sachdev and J. Ye, Gapless spin-fluid ground state in a random quantum heisenberg magnet, *Physical review letters* **70**, 3339 (1993).
 - [2] S. Sachdev, Holographic metals and the fractionalized fermi liquid, *Physical review letters* **105**, 151602 (2010).
 - [3] A. Kitaev, A simple model of quantum holography (part 1), <https://online.kitp.ucsb.edu/online/entangled15/kitaev/> (2015), accessed: 2025-04-12.
 - [4] A. Kitaev, A simple model of quantum holography (part 2), <https://online.kitp.ucsb.edu/online/entangled15/kitaev2/> (2015), accessed: 2025-04-12.
 - [5] S. Sachdev, Quantum statistical mechanics of the sachdev-ye-kitaev model and charged black holes, *International Journal of Modern Physics B* **38**, 2430003 (2024).
 - [6] D. A. Trunin, Pedagogical introduction to the sachdev-ye-kitaev model and two-dimensional dilaton gravity, *Physics-Uspekhi* **64**, 219 (2021).
 - [7] F. I. S. Massolo, Finite n sachdev-ye-kitaev model (2019), accessed: 2025-04-15.
 - [8] J. Maldacena and D. Stanford, Remarks on the sachdev-ye-kitaev model, *Physical Review D* **94**, 106002 (2016).
 - [9] S. W. Hawking, Black hole explosions?, *Nature* **248**, 30 (1974).
 - [10] S. Sachdev, Quantum spin glasses and sachdev-ye-kitaev models, arXiv preprint arXiv:2402.17824 (2024).